打分单位审计：Stage 4–13 与答辩后

2026-11-05 · 只读核对，每格的出处是脚本与行号（以代码为准）。三段分别由三个审计助手完成，原文件在 /scratch/zhang.zhiqi2/jlens/audit_20261105/。

Stage 4–9 · Stage 10–12 · Stage 13 与答辩后

2026-11-06 更新：本审计指出的打分单位问题，答辩后已重做的部分

以下数字都来自看结果前写定规则的重做项目（补充 9–14），复现已发表数字是每一项的硬检查。本页其余内容保持原样，后来被更新的说法在原处加了橙色标注，原数字未改。

- **X1–X4 改为全部作者（批次 1 第 1 项，已完成，硬检查 PASS）：** X1–X4 改为给全部作者打分（全部名字 token），并按作者排位分层。结论都不变：X1 仍与任务无关，L48 有一点小而确实的任务依赖（作者 − 年份 +0.024 [0.019, 0.030]）；X3 改写标题保留约 90% 的作者偏好，target 与 control 一样；X4 从 L8 起可解码出作者，各作者排位相同，认得出的论文方向上更高，但按预注册在测试集上不显著（L8 +0.023 [−0.001, +0.048]，v42_paper_vs_team.md §3）；但只靠标题文字的基线不弱于 Qwen，仍不能排除标题文字的解释。X2 用全部作者重新拟合作者子空间后，搬动量从 0.016 升到 0.032（完整替换的约 7%），但子空间本身仍几乎解码不出作者（留出 $R^2 \leqslant 0.009$），H3 仍证据不足。（381_stage13/reports/b1_xall_report.md：X1 节（测试集，作者 − 年份，团队聚类 bootstrap 2,000 次）；X3 behaviour 表（测试集 552 篇，恢复比 0.897 [0.868, 0.926]）；X4 节（L8，全部作者 +0.048 [0.039, 0.058]；Recognition 表）；X2 verdict 表（L48 R_auth）；New S_auth 表（authall_r2 ⩽ 0.009）；Hard check 表）
- **E4b 减 5 个假标题（实际每篇可用假标题个数见正文）（批次 1 第 4 项，已完成）：** E4b 逐个假标题排除版（偏离 D1，保留 4,654 篇，每篇平均约 1.5 个可用假标题，0–4 个；预注册版只减 false_0、1,395 篇时 L40 姓首 token 为 +0.071）：L40 姓首 token +0.076 [0.068, 0.083]，姓 − 首字母 +0.033 [0.023, 0.042]，逐个排除版（4,654 篇）第一个 token +0.066；原 E4b（1,395 篇）+0.061；置换对照首字母仍 +0.018，其余 ≈ 0；硬检查 E4b L40 +0.0609 PASS。标题位置 probe 的 target − control（同一引用档内）有三个版本：① 预注册版（只减 false_0；false_0 是唯一检查过标题复用的假标题，干净的单一对照；名 1,706、姓 1,395、首字母 1,590 篇）：名的第一个 token L24 +0.051 [0.007, 0.095]、L40 +0.049 [0.005, 0.095]、L48 +0.073 [0.027, 0.122]；姓 L24 −0.002 [−0.037, +0.036]、L40 +0.012 [−0.026, +0.048]、L48 +0.016 [−0.025, +0.056]；首字母 L24 +0.033 [0.005, 0.059]、L40 +0.031 [−0.001, +0.062]、L48 +0.056 [0.025, 0.089]（与补充 9 已发表值相同，e4b_alltok.md）。② 逐个排除版（事后偏离 D1；每篇平均约 1.5 个可用假标题；名 3,046、姓 4,654、首字母 4,812 篇）：名 L24 +0.037 [0.012, 0.062]、L40 +0.048 [0.021, 0.076]、L48 +0.051 [0.023, 0.081]；姓 L24 +0.019 [0.001, 0.038]、L40 +0.030 [0.013, 0.048]、L48 +0.042 [0.025, 0.060]；首字母 L24 +0.026 [0.012, 0.040]、L40 +0.014 [−0.002, +0.030]、L48 +0.031 [0.014, 0.048]。③ "5 个假标题平均"版（仅敏感性参考：论文同①，只按 false_0 剔除，false_1..4 中约

79% 本应剔除，含泄漏；名 1,706、姓 1,395、首字母 1,590 篇）：名 L40 +0.047 [0.012, 0.081]；姓 L40 +0.024 [−0.006, +0.052]、L48 +0.031 [0.001, 0.059]；首字母 L40 +0.021 [−0.002, +0.043]。结论：预注册版中名的第一个 token 在 L24–L48 为正，姓在 L24–L48 不显著（L40 +0.012 [−0.026, +0.048]）；事后的逐个排除版中姓从 L24 起为正（L40 +0.030 [0.013, 0.048]）。名的信号在三版中一致；姓的证据弱，只来自事后版本。首字母在个别层为正、模式不一致（未做多层校正），不作结论

（381_stage13/summary_b1_e4b5/perdonor_by_run_layer.csv（version = V0 / Vpd / V5x0，列 tc / tc_lo / tc_hi；引用档内 target − control，团队整群 bootstrap 2,000 次 95% CI）；381_stage13/reports/e4b_alltok.md；b1_e4b5_report.md）。

- **X7 五个假标题（批次 1 第 2 项，已完成）：** X7 补做了另外 4 个假标题，按 5 个平均判定，target、control 和全部论文都满足预注册判据（S(L52) 分别为 +0.022、+0.042、+0.028，均 ⩽ 0.05）；原来对照组只用 false_0 时为 0.051，属于单个假标题的波动。（381_stage13/reports/b1_x7f_report.md（target 593 / control 593 / 全部 1,186 篇，团队聚类 95% CI））
- **C-3b（补充 13，只作敏感性分析，含泄漏，不作结论）：** 5 个假标题版本不剔除标题复用：硬检查 PASS；L40 姓的第一个 token +0.077 [0.071, 0.084]，首字母 +0.034 [0.029, 0.039]，姓 − 首字母 +0.043 [0.035, 0.051]（6,841 篇）；target − control 的区间几乎处处在 0 以上，与 C-3 一样。（381_stage13/reports/c13_C3b.md（Primary contrasts at L40 表 V5x0_noexcl 列））
- **P1 / P2 用 5 个假标题（批次 1 第 3 项）：还在运行**，194 个计算单元已完成 187 个，结果尚未出来；结果经用户确认后补入。（g391_b1p12f 作业链（14_groups/b1_p12f_gpu_v2.sbatch））
- **答辩后已完成、以前写“没做 / 暂停”的项目：** X2 只替换子空间（补充 6，2026-11-05）：按预注册判据 H2 不支持、H3 通过，但作者子空间几乎解码不出作者，H3 证据仍不足；全部名字 token 的标题位置 probe（补充 9）：L40 姓的第一个 token +0.071 [0.058, 0.085]；X5 跨问法检索曲线：没有一层高于 L4；M5 能用已有数据回答的部分；CPU 项 C-1 到 C-5（Stage 7.1 加标题对照、Stage 8 CPU 代理、不剔除标题复用的敏感性、M1 换 4 个假标题、Stage 10 附加分析补齐全部已存层）。

（381_stage13/reports/x2_report.md；e4b_alltok.md 第 5 节；x5_retrieval.md；m5_report.md；c13_C1.md–c13_C5.md）

- **批次 2（补充 11、12）：** 代码、tiny 测试和硬检查的单元测试都已完成，批次 1 结束后由编排作业自动开始（含 Stage 9 Fit B 与 holdout、F1 对半拆分、名字位置 probe 等）；与预注册的出入记在出入记录 D12-1 到 D12-3（2-1、2-2 较省但含原始提示逐位复现重拟合的硬检查；2-11 重建标签、论文内 AUROC、两个基线、对照组只作描述），2-7 的来源改用不含年份的 X4 提示。**批次 3（补充 14）：** 3-1 Stage 10 补 L41–L47、L49，3-2 补 L0–L23：代码已就绪，第二批之后开始；3-3 Stage 7.1B 快路径：已停止（偏离 D14-1）：3-3 停止：只有 7 篇论文同时有旧逐前缀结果和快路径结果，不足以确认两者在 ±0.02 内等效（输出层 target−control 差值之差 +0.0059，按论文 Welch 95% CI [−0.0207, +0.0324]；论文级 bootstrap [−0.0005, +0.0181]，因每组仅 3–4 篇偏窄，不作判据）。差异来源为 bf16 舍入随前向长度变化；长度跨档比例 target 28%、control 15%。Stage 10 全程使用快路径，组间口径一致，一致性检查按 0.25 容差（351_stage10.py 第 919 行）通过率 99.81%。决定过程：原判据（⩾

> 99% 名次完全一致）未通过（79.8%）→ 用户指出事后判据的设计缺陷 → 在看到论文级结果之前写定并锁定修正判据 → 按修正判据（Welch 区间）仍未通过 → 停止；3-4、3-5（Stage 8 按 W 的 GPU 重算）不做。
>
> （381_stage13/config/prereg_addendum11.json、prereg_addendum12.json、prereg_addendum12_deviations.json、prereg_addendum14.json；381_stage13/reports/b2_groupA_handoff.md）

对照本审计：只打分第一作者的 X1–X4 已改为全部作者（上框第 1 条）；只减 false_0 的 X7、E4b 已改为 5 个假标题（第 2、3 条）；只打分第一个 token 的 E4b 已由补充 9 扩展到全部名字 token；Stage 5–7 加标题对照见 C-1；Stage 8 的 W 只做了 CPU 代理（C-2），GPU 重算不做。（PROGRESS.md 2026-11-06 条；381_stage13/config/prereg_addendum10.json、prereg_addendum13.json）

要点（主会话汇总）

- **只给第一作者打分**：Stage 5、7.2、7.3、7.4、8（W）只用第一作者的第一个 token；Stage 13 的 X1、X2、X3、X4 只用第一作者（但给该作者的全部名字 token 打分）。Stage 10–12 没有只看第一作者的。
- **只给首 token 打分**：Stage 4 主指标、5、7.2、7.3、8 的 W；Stage 11 E4b（及其变体）。Stage 8 的 200 篇里 129 篇（64.5%）第一作者的首 token 是首字母，所以 W 主要测的是首字母。
- **只用 false_0**：E1（已由 X6 用 false_1..4 补过）、E2（CPU 已补：L0 损失 0.831→0.820，结论不变）、P1、P2（需 GPU 约 9 A100 时，建议写局限）、E4b 和全 token probe、X7（对照组 S(L52) 0.051 只比阈值高 0.001，补 false_1..4 约 1.3 A100 时）。
- **没有对照**：Stage 4 条件 MRR、Stage 6 JMRR、Stage 7.1 只看真标题；可用 Stage 10 已存数据在 CPU 上改成标题特异对比（Pilot50 和 Stage 8 的论文都在 Stage 10 里）。
- **不覆盖全部论文**：干预实验用 1,186 篇样本；E4b 确认版去掉标题复用后只剩 1,395 篇（80% 被去掉，全 token 版同）；C/D、R2 未写作者组样本小，只作描述。
- **层**：没有分析用全部 64 层（5–17 层不等）；Stage 10 不对"起始层"下结论。
- **M5**：CPU 完成（事后）：名字位置真假标题区分 输出 0.919，中间层 lens 0.51–0.56；标题和 ":" 位置 probe AUROC 约 0.50。已在 v28 第五部分；STATUS 未更新。**V42**：CPU 完成：论文 vs 团队重抽区间宽度几乎相同（中位比 0.97–1.00），只有 2 行"是否含 0"改变；已在 v28。
- **文档待改**：v28 里有 4 处仍写"全 token probe 没有结果"；STATUS 仍写 X2 未做；E4b"目标 > 对照"这句应写明主要来自首字母和名（姓首 token 到 L56 才为正）。

Part A — Scoring-unit audit, Stages 4–9 (read-only, 2026-11-05)

Code paths are relative to `~/jlens_work/`; outputs relative to `/scratch/zhang.zhiqi2/jlens/`.

Line numbers are from the current files (byte-identical to the frozen originals). Everything below was checked in the code; the methods handbook (`351_stage10/reports/methods/parts/C_stage1_9.html`) was used only as a map.

Shared facts (all of Stages 4–8):

- The prompt is `Who are the authors of the paper titled "{title}" ({year})?\nAuthors:`. The readout position is the last token. The name is tokenised on its own as `encode(" "+name)` and appended as token IDs (`04_stage4/351_stage4_readout.py:49-61`, `05_stage5/351_stage5.py:248`).
- The readout helper is `05_stage5/351_stage5.py:191-208` (`readout`). Each context gives 15 modes: public J-lens (Neuronpedia, 557 prompts, sha `0af5ccf2…`) at L16/24/32/40/48/56/62, the vanilla logit lens at the same 7 layers, and `final_L63` (`351_stage5.py:8, 203-208`). Stages 6, 7.2, 7.3 and 8C all import this helper.
- "First author" always means the first name in the `;`-separated author string: `first_author()` (`351_stage5.py:35-37`), `real[0]` (`08_stage8/351_stage8B.py:254`), and `unique_names(r)[0]` in Stage 4 (`351_stage4_readout.py:471`).
- **New finding (CPU count, this audit):** the gold first author's **first token is just an initial** (e.g. "M. Ausloos" → first token " M", id 386) for **129/200 (64.5%) of Stage 8 papers** and **31/50 (62%) of Pilot50 papers**. For the false names it is 14,376/20,000 (Stage 8) and 3,460/5,000 (7.2). This is a regex count on `351_stage8/controls/stage8_author_candidates.csv` and `351_stage7_2/candidates/pilot50_author_candidates.csv`. The mean gold name length is 4.5 tokens (Stage 8) and 4.3 tokens (Pilot50). The 100 false names have on average only 43.95 (Stage 8) and 45.76 (7.2) distinct first tokens per paper. So every first-token W mostly measures whether the gold *initial letter* beats about 44 other initials or given-name starts.

(1) Scoring-unit table

#	Experiment	1 Authors scored	2 Tokens scored	3 Comparison	4 Layers
4a	**Stage 4 primary** (`first_list`	**First author only** (`is_lead`, `04_stage4/3`	**First token only** (`first_rr=1/ite`	**None.** Absolute RR = 1/rank against the whole	7: L16/24/32/40/48/56/62 + final L63 (:384, :186)

#	Experiment	1 Authors scored	2 Tokens scored	3 Comparison	4 Layers
	`ed_first_rr`)	`51_stage4_readout.py:120-121, 173, 188`)	`ms[0]['rank']` , :469)	vocabulary. No false name and no false title (`summary.json` "control_analysis: NOT RUN", :238)	
4b	Stage 4 auxiliary `all_author_first_rr`	All authors (93, deduplicated by casefold, :42-47, :453)	First token only (:192)	None	Same 7 + L63
4c	Stage 4 auxiliary `conditional_token_mrr`	All authors	**All name tokens**, teacher-forced (:458-469)	None	Same 7 + L63
5	**Stage 5** crossed title × author	**First author only** (G = first author of P; F = first author of donor Q) (`05_stage5/351_stage5.py:108-121`)	**First token only** (`vec[gold[0]]` , `vec[false[0]]` , :248, :260-262)	**One false name (F) + one false title (Q's title)**, crossed. T = (logit G1 − logit F1 \	true title) − (same \
6	**Stage 6** conditional JMRR	**All authors** (227 author instances, no dedup, `06_stage6/351_stage6.py:45, 176`)	**All name tokens**, teacher-forced (972 tokens, :179-190). JMRR = mean_token(1/r), then mean over authors, then over papers (:222-231)	**None.** True title only. Absolute RR vs vocabulary. A–B is the only contrast (:287-305)	7 + L63; integrated = mean of 7 (:239-242)
7.1	**Stage 7.1** first vs continuation	All authors	Decomposed into **first token**, **continuation tokens** (multi-token authors only, no zero fill) and **all tokens** (`07_1_stage7_1/351_stage7.py:80-90`)	None (true title only), A–B pairs (:108ff)	7 + L63
7.1B	Stage 7.1B terminal	All 227 authors	All name tokens (every prefix)	True + 20 metadata-matched false	7 + L63

#	Experiment	1 Authors scored	2 Tokens scored	3 Comparison	4 Layers
				titles (the 7.3 set) (`351_stage7_1B_terminal.py:100, 158, 266, 292`)	
7.2	**Stage 7.2** gold vs 100 false names	**First author only** (`p['first_author']`, `07_2_stage7_2/351_stage7_2.py:80, 75`)	**First token only** (`first_token_id`, :115; raw stores only the logits of the unique candidate first-token IDs, `351_stage7_2_score.py:120-124`)	**100 false names** (50 matched + 50 random from the 6,841-paper author universe) under the **true title only**. W = mean over 100 of [1(g>f) + 0.5 • 1(g=f)] (:129-130, :150-158). Ties count 0.5	7 + L63
7.3	**Stage 7.3** shuffled-title null	First author only (gold = 7.2 gold, `07_3_stage7_3/351_stage7_3.py:84`)	First token only (`351_stage7_3_score.py:144, 169`)	Same 100 false names in each context, plus **20 metadata-matched false titles**. TA = W(true) − mean of the 20 W(false) (:173-186)	7 + L63; integrated = mean of 7 (:203-208)
7.4	Stage 7.4 cross-stage audit	Inherits: first author	Inherits: first token	Stage 5 single-cross logit T vs 7.3 TA (`07_4_stage7_4/351_stage7_4.py:39-49`); J-lens − vanilla TA (:51-66); low-bin L56 (:68-77)	7 (correlation per layer; floor at **L56 only**)
8A/8B	Stage 8 freeze	Gold = **first author only** (`08_stage8/351_stage8B.py:254`)	First-token IDs frozen (:446-454)	50 matched + 50 hash-ordered "random" false names (:360, :372); 20 false titles (:505-540 per handbook)	—
8C	Stage 8C scoring	First author	**First token only**: per paper the sorted set of 101 candidate first-	21 contexts (true + 20 false titles)	7 + L63 (`:12-14`)

#	Experiment	1 Authors scored	2 Tokens scored	3 Comparison	4 Layers	
			token IDs (`351_stage8C.py:126`); logits of those IDs only (:374-382). No continuation tokens			
8D-RQ1	**8D RQ1**	**First author only**	**First token only**	W(true) − mean of 20 W(false) on the **combined 100 false names** (`351_stage8D.py:51-54, 186-224`)	**L56 only** (:243-245)	
8D-RQ2	**8D RQ2**	First author	First token	Paired TA(A) − TA(B) (:226-249)	**L56 only**	
8D-RQ3	8D RQ3	First author	First token	Low − high band, pair-balanced (:251-270)	**L56 only**	
8D-JV	8D J − V	First author	First token	TA(J-lens) − TA(vanilla) per paper (:283-291)	**L56 only**	
8D-sens	8D sensitivities	First author	First token	Excluding gold-tied false names; excluding source-overlap titles; titles with token gap ⩽ 10 (:196-224)	All 15 modes stored; reported at L56	
8Dsup	**8D supplement** (frozen primary endpoint)	First author	First token	Same TA, read from 8D `paper_layer_advantage.csv` (`351_stage8D_supplement.py:45-57`)	**Mean of the 7 layers** (:16, :52); per-layer A–B descriptive (:58-59)	
8E-rc	8E / Holm / LOPO recompute (2026-10-30)	First author	First token (gold id from candidates, `recompute_2026-10-30/stage8_frozen_plan_recompute.py:208`)	4 weightings of the 100 false names (author / unique / no-gold-tie / unique-no-tie); Holm families over 7 layers (:137-147)	L56 for weightings; 7 layers for Holm	
9	**Stage 9 FullFit**	**No authors are scored.**	n/a (WikiText tokens 16…	n/a. 9B hold-out comparison was	Source L16/24/32/40/48/56/62	

#	Experiment	1 Authors scored	2 Tokens scored	3 Comparison	4 Layers
		Lens fitting only	seq−2, skip_first 16, max_seq_len 128, `09_stage9/stage9_a_repair.py:152`)	**never implemented**	→ target L63 (`351_stage9.py:10`)

Direct answers to the two "must check" items

- **Stage 8 W** uses the **first author only** (`real[0]`, 8B:254) and the **first token only** (8C:126, :380-382; 8D:51-54, :190). The comparison is against the first tokens of 100 false names (≈ 44 distinct tokens). Ties (673 gold–false first-token collisions) count 0.5. The primary statistics in 8D are at **L56 only**. The frozen-plan endpoint (the 7-layer mean) appears only in the supplement.
- **Stage 4–7 RR / JMRR**:
- Stage 4 *primary*: first author, first token, no comparison.
- Stage 4 conditional MRR, Stage 6 JMRR and Stage 7.1 "all": all authors and all tokens, teacher-forced, true title only, no comparison.
- Stage 7.1 splits these into first vs continuation tokens. The A−B difference sits in the **continuation** tokens (+0.0206 [+0.0022, +0.0385]); the first token gives +0.0014 (n.s.).
- Stages 5, 7.2, 7.3, 7.4 and 8: **first author × first token** only.

(2) Restriction analysis

Direction is relative to the reported conclusion (larger = the restriction inflates the reported effect; smaller = the restriction hides effect).

CPU-fix source: Stage 10 raw (`351_stage10/raw/<cond>/<readout>/L*/shard_*.npz`) stores, for **every author and every name token** of all 9,108 papers (this includes all Pilot50 and all 200 Stage 8 papers, checked: 50/50 and 200/200, 596 Stage 8 author contexts), the gold-token `rank`, `ties`, `logprob` and `top1`.

- Conditions: true, false_0..4 (Stage 10's own 5 matched false titles, **not** the 7.3/8B titles) and delete.
- Readouts: logit and public J-lens at L24/32/40/48/50–62, self100 and fitA at 6 layers, and final.
- **Not stored:** L16, any false-author score, or any full-vocabulary vector.
- Stage 10 tokenisation equals Stage 6 (972/972 identical IDs, `351_stage10/reports/crosscheck_stage6.csv`).

GPU cost basis: the Stage 10 fast path did 236,677 author-contexts × all readouts in about 9 A100-GPU-h, i.e. about 0.14 s per author-context. The old per-prefix path (Stages 4–8) runs at about 6 s per context.

Experiment / cell	Restriction	a. Effect on conclusion (reason)	b. Fixable on CPU from saved dat cost
4 primary: authors	First author only	Unclear: first authors may be better or worse known than co-authors. Pilot all-author first-token RR correlates r ≈ 0.69–0.77 with the first-author one	**CPU yes:** `351_all/351_stage4_artifacts/ _scores.csv` has all 93 authors × × 15 modes (5,985 rows; columns reciprocal_rank, logprob); `all_author_first_rr` is already `summary/paper_metrics.csv`
4 primary: tokens	First token only	**Smaller.** About 62% of first tokens are an initial; Stage 10 shows initials carry little (+0.11) vs later tokens (+0.45)	CPU yes (same file, `conditional_token_mrr`)
4: comparison	None (absolute RR)	**Larger / uninterpretable:** an absolute RR mixes name guessability with title-specific knowledge. Stage 5 was created for this reason	Not fixable from Stage 4 data. Stag gives true − false_k − delete for the papers on CPU
4: sample	20 papers, unmatched; L56 mean driven by one paper (71% of sum)	Larger at L56 (one outlier)	CPU: Stage 10 covers all 9,108 pap
4/5/6/7: layers	7 layers (L16…62)	Unclear for the mean; **misses the L50–61 ramp** that Stage 10 found. Coarse layers cannot locate the onset	CPU partially: Stage 10 raw has L5 public/logit for the same papers (r GPU re-run of the 7.2/7.3/8C first-t contexts at all 63 lens layers needs 38–67 stored IDs per context: 1,05 contexts, ≈ 1–2 GPU-h with a batc (≈ 9 GPU-h with the old 6 s/conte
5: authors	First author only (gold and false)	Unclear	Gold side: CPU via Stage 10 (all au vs 5 false titles). False-name side:
5: tokens	First token only	**Smaller** (initials; 7.1 shows the A–B signal is in continuation tokens)	Raw `351_stage5_crossed_v2/raw/pil lts.jsonl` stores only gold/false t logit and rank: **not** fixable on CPU crossed design. GPU full-name cro run: 100 contexts × 2 names ≈ < ((fast path)
5: comparison	**One** false name + **one** false title	Unclear, high variance. A single donor can push T either way (7.4: r = 0.45 with the 20-title version)	Raw cannot add references. Stage gives 5 false titles (gold only)
5/6/7.x: sample	Pilot50 (50 papers, 20 pairs;	Larger at L56 (winner's curse); otherwise unclear (low power)	CPU via Stage 10 for the gold-rank the full cohort

Experiment / cell	Restriction	a. Effect on conclusion (reason)	b. Fixable on CPU from saved dat cost
	L56 chosen after inspecting it)		
6 / 7.1: comparison	None (true title only, absolute JMRR)	**Larger**: the A–B difference may be name guessability rather than title knowledge. In Stage 10 the true-title-only target − control (+0.247) exceeds the title-specific one (+0.207)	**CPU yes**: Stage 10 raw has the sar authors × 972 tokens under true a titles (and delete) at L24–62 for pu logit. Pilot50 A/B title-specific JM recomputed (L16 missing)
6 / 7.1: tokens	All tokens, teacher-forced	Continuation tokens are conditioned on the correct prefix → larger than free recall	n/a (a design property)
7.1B: completeness	7.5% of contexts, no analysis	No conclusion exists	Finishing needs about 14,353 con ≈ **24 GPU-h** (old path), or about 1 re-implemented on the fast path. equivalent question is answered c Stage 10 (all tokens × 5 false title
7.2: authors	First author only	Unclear	False-name logits exist only for fir IDs → full name or co-authors nee Gold side only: CPU via Stage 10
7.2: tokens	First token only (62% initials; 100 names ≈ 46 distinct tokens)	**Smaller** for the knowledge effect: W is mainly initial-letter discrimination; 3.1% ties at 0.5 shrink W toward 0.5. The tie sensitivity already exists	Raw stores logits only of the uniqu token IDs (67 for the first paper) → needed for full-name W: 50 × 101 5,050 author-contexts ≈ 0.2 GPU- (≈ 1–2 h with a new script incl. lo prefix old path ≈ 15k contexts ≈
7.2: comparison	100 false names, **true title only**	Larger (name prior not removed); 7.3 fixes this	—
7.3: authors / tokens	First author, first token	**Smaller** (same reasons as 7.2)	GPU (≈ 1,050 contexts × 101 nam author-contexts ≈ **4 GPU-h** fast p Partial CPU proxy: Stage 10 gold a title-specific rank for the same 50 different false titles, rank instead
7.3: comparison	20 metadata-matched false titles (not exchangeable)	Unclear; tail fractions are not p-values (already stated)	—
7.4	Inherits first author/token; floor result at L56 on 6 papers	Larger (6 papers, L56 pre-inspected)	CPU: Stage 10 full-cohort bin 0-2
8D RQ1/RQ2/RQ3/J–V:	**First token only** (64.5% initials;	**Smaller** for a full-name knowledge effect: an initial	**Not on CPU** for W: `351_stage8/raw/stage8C_contex`

Experiment / cell	Restriction	a. Effect on conclusion (reason)	b. Fixable on CPU from saved dat cost
tokens	≈ 44 distinct false tokens)	is weak evidence (Stage 10: initial +0.11 vs later tokens +0.45; Pilot 7.1: A–B lives in the continuation tokens). But the RQ2 point estimate is fragile anyway (it changes sign after removing 10 pairs), so the *direction of RQ2* under full names is **unclear**	`jsonl` (4,200 rows) stores `score_` only for the per-paper set of candi **token** IDs (38 for the first paper); r continuation tokens or full vocab name W re-run: 4,200 contexts × ≈ 424k author-contexts ≈ **16–20 80GB GPU-h** on a Stage 10-style f (new script = design change, need approval); ≈ 300+ GPU-h with the prefix path. **CPU proxy available** Stage 10 raw for the same 200 pap pairs (596 author contexts, all tok vs false_0..4, L24–62 public/logit, title-specific log10(1/rank) for firs all tokens and for all authors, pair minutes-long CPU job, but a differ and different false titles (cannot re
8D: authors	First author only	Unclear	CPU proxy as above (Stage 10, all a for co-authors needs GPU (×3 the above; 596 vs 200 author contexts
8D: comparison	100 false names, collapsed to ≈ 44 tokens; ties 0.5	Small either way: unique-token weighting gives RQ1 +4.58 vs +5.18 (smaller) and RQ2 +5.40 vs +4.98 (larger); no-tie gives +5.32 / +5.03 (already recomputed, `351_stage8/recompute_2026-10-30/s8_8E_weighting_by_paper.csv`)	Done on CPU
8D: comparison (titles)	20 matched false titles; title-token gap mean 7.37; slope +0.12 pp/token	Slightly larger (longer-gap titles give a larger TA); short-title sensitivity +4.79 [−0.24, +9.83] for RQ2	Done on CPU
8D: layers	**L56 only** for RQ1–RQ3 and J–V (frozen plan said 7-layer mean)	**Larger**: L56 was picked on the pilot; the 7-layer mean is about half (+2.33 vs +5.18 pp RQ1; +2.41 vs +4.98 RQ2). L56 also sits on the steep L50–61 ramp, so a one-layer endpoint is sensitive to the exact layer	Done on CPU (supplement + Holm recompute: integrated RQ2 surviv RQ2 does not survive the 7-layer H
8D: sample	200 papers, A/B only (R0), 100 exact-matched pairs (biased to	Unclear (selection toward few-author papers)	CPU: Stage 10 full cohort (target/c

Experiment / cell	Restriction	a. Effect on conclusion (reason)	b. Fixable on CPU from saved dat cost
	easy-to-match cells)		
8D RQ3	25 + 25 pairs, L56	Unclear; underpowered (CI −3.69 to +8.60)	CPU (integrated version already +(+3.18])
9: scoring	Nothing scored; Fit B 66/100; 9B/9C not implemented	No Stage 9 conclusion about stability exists	Finishing Fit B: 34 prompts × ≈ 3 **GPU-h** (checkpoint intact, resum load; 9B comparison ≈ < 1 GPU-h

Bottom line

- The biggest unmeasured restriction is **first author × first token** in every win-rate experiment (Stages 5, 7.2, 7.3, 8). In about 64% of the papers that token is a single initial letter.
- This cannot be fixed from saved Stage 7/8 data, because only the candidate first-token logits were stored.
- Two ways to address it:
- a cheap **CPU proxy** from Stage 10 raw (same papers, all authors and tokens, rank metric, different false titles);
- a **GPU full-name W redo**: about 16–20 A100-GPU-h for Stage 8, plus about 4 for 7.3. This needs a new script and user approval.
- The layer restriction (L56 only) is already handled by the 2026-10-30 recompute.

(3) Could not verify

- The 8B false-title tier rules (`351_stage8B.py` ≈ 505–540) and the 7.3 `tier()` function were not read line by line; the line ranges are taken from the handbook. The selection of 8B matched/random names was confirmed only at the `[:50]` cuts (:360, :372).
- The 7.1B task builder was skimmed (grep), not read in full. The authors × tokens × 21 titles structure is inferred from :92-292 and the 15,519/227 asserts.
- The GPU-hour figures are estimates, scaled from the Stage 10 main round (about 9 GPU-h / 236,677 author-contexts, PROGRESS.md) and from the 6 s/context of the old path. No timing test was run.
- The initial-first counts use a regex on the first word (`[A-Z].?`-style). Hyphenated or unusual initials may be slightly miscounted.
- Whether Stage 10's per-token data for the Stage 8 papers exactly reproduce Stage 8's gold first-token rank was not checked. Stage 10 recomputes with the fast path, and the parity gap is ⩽ 0.5 logit.
- Stage 9 scripts were checked only for layers and fitting settings (grep), not end to end.

Audit part B: what each Stage 10–12 readout scores

Written 2026-11-05. Read-only audit. No file was edited and no job was submitted. Every cell below was checked in the code. Line numbers refer to the current files:

- `~/jlens_work/10_stage10/351_stage10.py` (S10)
- `stage10_extra.py` (X), `stage10_extra_format.py` (XF), `stage10_extra_cond_role.py` (XC), `stage10_extra_r0.py` (R0), `stage10_extra_r0r1.py` (R01)
- `~/jlens_work/11_stage11/361_stage11.py` (S11), `stage11_analyze.py` (A11)
- `~/jlens_work/12_stage12/371_stage12.py` (S12), `stage12_analyze.py` (A12), `stage12_cpu.py` (C12)

Result paths are relative to `/scratch/zhang.zhiqi2/jlens/`. The methods handbook (`351_stage10/reports/methods/parts/D_stage10_11.html`, `E_stage12.html`) was used only as a map. Where it agrees with the code, this is noted as "handbook agrees".

0. Conventions every Stage 10–12 per-token score shares

- **Context.** Each author is scored alone: `PROMPT + " " + author`, teacher-forced. Token j is scored at the position just before it (S10:118–136, 563–564).
- **Authors.** Every author in the label file is kept. The `authors` column is split on ";" (S10:118). Authors are excluded only if their name has more than 32 tokens or the context has more than 512 tokens. There were 0 exclusions (`351_stage10/config/manifest.json`: n_excluded_authors 0). The cohort has 33,811 author contexts and 149,999 name tokens.
- **Paper value.** For each paper, the paper value is the token-weighted mean over all name tokens of all authors of nlr = log10(1/rank) (S10:651–656, `mean_nlr`). The per-author mean `author_mean_nlr` (S10:657) and role-specific means (S10:659–662) are secondary.
- **Deviation from the frozen protocol.** `protocol.json` names the primary metric as "sum of 1/rank" (S10:165). Every analysis instead uses the mean log10(1/rank). This choice of metric does not restrict which tokens are scored, but it should be stated in the methods.
- **Ranks.** Rank is over the full 151k vocabulary, ties counted favourably (S10:442–449). The only exception is the probes (E4b, X1), which rank within a 4,314-id name vocabulary.

1. Main table: one row per experiment or readout

Column abbreviations:

- **Authors.** all = every scored author of the paper.
- **Tokens.** all = every name token. "first" = token_index 0 of each author (an initial in about 73% of authors). "surname-first" = the first token of the last word.

- **Comparison.** mean5 = true minus the mean of false_0..4. f0 = false_0 only, the best-matched donor (S10:303 `k` = cumcount after sorting by tier and cost).
- **Readout.** L63/final = model output. logit = logit lens. public = public J-lens (Salesforce-wikitext). self100 and fitA are fitted J-lenses. alens = author-prompt lens (F1).

#	Experiment / readout	1 Authors	2 Tokens	3 Comparison	4 Layers	5 Sa
10.1	Stage 10 true-title level, groups A–E (`analyze`)	all (S10:118)	all; role means as secondary (S10:651–662)	none: true title only; target vs control within citation bin, bins with ⩾30 per group (S10:670–692, 752)	public and logit: 24, 32, 40, 48, 50–62 (17 layers). self100 and fitA: 24, 32, 40, 48, 56, 62. final: 63 (S10:42; `raw/*/L*`, counted)	all 9 run. uses targ con ⩾30 R0/R with mc_ (S10 uses ⩾12
10.2	Title-specific signal (spec), total, topic (`analyze-title`)	all	all; also surname-first, initial-first, punct and other role metrics (S10:834–835)	**mean5** (S10:857, 863–865). Also `spec_strict`: mean over strict-tier donors only (S10:869)	same as 10.1; the extension layers L50–61 are included because `load_raw` picks up every complete (readout, layer) (`stage10_ext_analyze.sbatch`)	all 9 con 3,73
10.3	Conditions true / false_0..4 / delete (raw runs)	all	all	7 conditions, each scored separately	as 10.1, for every condition (72 shards each, counted)	all 9
10.4	R0-only sensitivity	all	all; surname-first column as secondary	mean5 (reads `paper_title_contrasts`)	every readout and layer in 10.2	R0 p 3,42 2,15 fina agre with uses cell
10.5	R0/R1 views (A−B, C−B, C−A, subgroup means)	all	all	mean5	every readout and layer	targ spli leve C−A

#	Experiment / readout	1 Authors	2 Tokens	3 Comparison	4 Layers	5 Sa
						per (R01
10.6	R1 per-author, written vs missed	**per author**, paired within paper; also a version without the first author (R01:150–151)	all	mean5 (R01:144)	every readout and layer	R1 targ pap rebu cou m, k only both and auth pap with first
10.7	Fine layers (extension)	all	all	mean5	public and logit at L50–55 and L57–61	all 9
10.8	X1 token / word position	all	**split** into first token vs later tokens, and first word vs later words (X:95–97)	mean5 (X:66)	**6 main layers + 63 only** (X:26 PLAN = DEFAULT_LAYERS)	all 9
10.8b	X1 by name format	all, split by the role of the author's first token (initial_first / given_first)	first token vs later tokens	mean5 (XF:58)	**5 readout-layers only**: public 56 and 62, logit 56 and 62, final 63 (XF:12)	all
10.8c	X1 conditional by first role	all	later words, conditioned on the first word being in the top 10	none (true title only)	**3 readout-layers**: final 63, logit 62, public 62 (XC:87)	all
10.9	X2 citation bins and Spearman	all	all	mean5	6 main layers + 63 (X:26)	all, × b
10.10	X3 lens comparison	all	all	mean5	6 main layers (X:164)	all

#	Experiment / readout	1 Authors	2 Tokens	3 Comparison	4 Layers	5 Sa
11.1	E1 tail + name patching	all authors of each sample paper (S11:239)	all (rows = Stage 10 row_ids, 19,077 tokens)	**f0 only** (S11:57, 236; numerator and denominator both use the false_0 run, A11:64–66)	24, 32, 40, 48, 50, 52, 54, 56–63 (S11:52)	Stag sam targ con pap per bin, (S11 `con ol.`
11.2	E1t last-title-token patching	all	all	**f0 only**; the baseline is the L63 element of the same batch (A11:56–60)	8, 16, 24, 32, 40, 44, 48, 52, 56, 60, 62, 63 (S11:58)	1,18
11.3	E2 knockout (after-title positions → title)	all	all	The knockout is applied to the **true run only**. The **denominator alone** uses f0: true − false_0 from E1 (A11:87, 91, 94)	starting layers 0, 24, 32, 40, 48, 50, 52, 54, 56, 58, 60, 62; baseline 64 (S11:53)	1,18
11.4	E2n knockout (":" + name positions only)	all	all	as E2 (denominator f0)	as E2	1,18
11.5	E3 author chain	all, joined in one sequence (S11:286–292)	all	mean5 (A11:243–246)	final 63; logit and public at 40, 48, 52, 56, 58, 60, 62 (S11:54–55)	all 9 con unit
11.6	E4 title-position lens readout	all	**first token** + **surname-first token** only (S11:366–367). The report table shows only `first` (A11:140). The CSV also has `surname` rows (A11:126)	mean5 (A11:122)	logit and public at 16, 24, 32, 40, 44, 48, 52, 56, 58, 60, 62; final 63 (S11:59–60)	all 9 con + fal dele
11.7	E4b title-position probe	all (mean over authors	**first token only** (A11:163: `gold.kind ==`	**f0 only**. Residuals were	16, 24, 32, 40, 44, 48, 52, 56, 60	trai test

#	Experiment / readout	1 Authors	2 Tokens	3 Comparison	4 Layers	5 Sa
		of the paper)	`'first'`; same in R01:233)	stored only for true and false_0 (S11:358). spec = probe(true) − probe(false_0) (A11:189–191)	(S11:61)	9,10 pap con bin. con vers fold reus drop targ con
11.7c	E4b controls (X1 in stage12_cpu)	all	**first token only** (C12:120)	**f0 only** (C12:118). Controls: gold permuted to any other paper / to a paper of the same field (C12:123–127, 150–151); probe trained on false-title pairs (C12:142–148)	**only 24, 40, 56** (C12:137)	all 9
11.8	E1/E2 by recall level (R0, R0/R1)	all	all	as E1/E2 (f0)	as E1/E2	R0 s 1,18 and view
12.1	P1 single-position patching	all	all (19,077)	**f0 only**: the baseline is `none`, an unpatched false_0 element in the same batch (S12:101, 119; A12:48, 55)	32, 40, 44, 48, 50, 52, 56 (S12:35)	1,18

#	Experiment / readout	1 Authors	2 Tokens	3 Comparison	4 Layers	5 Sa
12.2	P2 attention vs MLP	all	all	**f0 only** (`none`, same batch) (S12:146; A12:69–73)	8, 16, 24, 28, 32, 36, 40, 44, 48, 52, 56 (S12:37)	1,18
12.3	N1 multiple-choice name prior	all authors of every option list. Option 0 has the same number of authors as the paper for 9,108/9,108 papers (checked)	all tokens of the option lists (`mean_nlr`, S12:205–206). Names are in initial_surname format, not the Stage 10 spelling. first_author_nlr and first_token_nlr are stored but not analysed	true list vs **mean of the 3 distractor lists** (margin) and vs their max (A12:88). No false-title comparison	L63 only (S12:203)	9,10 (36, row kee 8,89 List thar are (S12
12.4	N2 pure name prior	all	all	pure vs delete vs true (no false title)	stage12_analyze: final 63 only (A12:109). Stored and analysed post hoc in Stage 13: logit and public at 48, 56, 62 (S12:39; `13_stage13/stage12_n2_layers.py`)	all 9
12.5	F1 author-lens fit	full author list in the fit text (S12:56)	all positions after skip_first = 16	–	sources 24, 32, 40, 44, 48, 52, 56, 62; target 63 (S12:40, 233)	**40 p** outs Stag sam targ con (S12
12.6	F2 author-lens readout	all	all, at the name positions; at the title position, all name tokens are ranked against the title-position logits (S12:263–264)	mean5 (A12:140)	the 8 F1 layers	Stag sam pap shar nam fit p 1,18 139)

#	Experiment / readout	1 Authors	2 Tokens	3 Comparison	4 Layers	5 Sa
12.7	stage12_cpu A: prior / topic / spec decomposition	all	all + roles (C12:12, 41–43)	mean5 (`false_*` columns of `paper_title_contrasts` = mean5)	every readout and layer in 10.2	all 9
12.8	stage12_cpu B: prior vs recognition	all	all	mean5 (spec); delete (prior)	**final 63 only** (C12:67)	R0/F mc_ (C12
12.9	stage12_cpu C: surname commonness	all (surname = last word of each author, C12:89–91)	all	mean5	final 63 only	all

2. Restricted cells: direction of the effect, CPU or GPU fix, recommendation

GPU cost basis: per-unit compute seconds come from the job logs (`grep 'stage11 <unit> … shard … Ns'` and the matching `stage12` lines). Totals over the 1,186-paper sample:

Unit	Compute time
E1 patch	1,741 s
E1t	1,582 s
E2	908 s
E2n	907 s
titlepos (all 9,108 papers × 6 conditions)	6,402 s
P1	5,464 s
P2	1,700 s
Stage 10 main round	504 shard-conditions × about 45

Each additional job also needs about 0.3–0.5 h for loading the model (prefetch plus load). All costs below are in A100-80GB GPU hours.

Restricted cell	a. Effect on the conclusion	b. Fix: CPU from saved or GPU (A100-h)
E2 / E2n: denominator uses f0 only	Smaller, but negligible. false_0 is the best-matched donor, so true − false_0 is a slightly smaller gap than the gap to the mean of five. **Computed in this audit on CPU**: the gap is 0.316 with false_0 vs 0.320 with the mean of five (all papers); target 0.418 vs 0.426; control 0.214 vs 0.214. E2 loss from L0: 0.831 → 0.820 (all), 0.839 → 0.822 (target), control unchanged. L24 0.712 → 0.702, L32 0.372 → 0.367, L48 0.064 → 0.063. E2n at L0: 0.153 → 0.151.	**CPU, done.** The numer never involves a false ti The E1 true and false_0 equal the Stage 10 final ranks on 100% of the 19 rows, so Stage 10 `cond_false_1..4/fina` gives an exact mean-of- denominator.
E1: f0 only	Direction unclear in principle. Already checked: Stage 13 X6 reran E1 with false_1..4. Target L48 is 0.788–0.803 across the five titles (average 0.794); L52 is 0.984–0.990 (`381_stage13/reports/x6_report.md`).	Needed GPU. This is alr done (X6).
E1t: f0 only	Direction unclear. Recovery is small (⩽0.15 all), so changing the donor is unlikely to change the "little at the title token" reading.	GPU: 4 × 1,582 s ≈ 1.8 load ≈ **2.2 A100-h**. Not fixable: the patched run only for false_0.
P1: f0 only	Direction unclear. X6 shows the full-tail E1 curve does not depend on which false title is used, so the same position profile is likely.	GPU: 4 × 5,464 s ≈ 6.1 load ≈ **6.5–7 A100-h**.
P2: f0 only	Direction unclear, same argument as P1.	GPU: 4 × 1,700 s ≈ 1.9 load ≈ **2.3 A100-h**.
E4b: first token only	**Understates the effect.** The scored tokens are mostly initials, which carry the weakest title-specific signal (10.8, 10.8b). Stage 13 addendum 9 (`381_stage13/reports/e4b_alltok.md`, written 2026-11-05 17:38) scored all tokens by role. The all-paper signal is >0 for every role and layer (given_first up to +0.163, surname_first up to +0.098). Target − control is positive for initial, given_first and given_cont at most layers. **For surname_first, target − control is ≈ 0 at L24–L52 and only positive at L56 (+0.058) and L60 (+0.043).** So the target − control part of the E4b claim rests on the first-name and initial tokens. The surname tokens show it only late.	Needed a GPU data step states at every name-predicting position). Th already done in addend A CPU-only "surname-f variant from the saved data is possible (the go `surname` and its vocab are stored), but its targ L63 state at ":", predicts 0, not the surname, so i a clean fix.
E4b: f0 only	Probably understated, though the direction is not certain. false_0 is the most similar donor (same tier, lowest cost), so probe(false_0) is closest to probe(true). The mean of five false titles would likely give a larger spec. The title-reuse exclusion would then need to cover five donors, leaving fewer papers.	GPU. Residuals for false were never stored (S11: Rerunning titlepos with residuals only, 4 condit 72 shards × about 15 s + load ≈ **1.5–2 A100-h** CPU probes.

Restricted cell	a. Effect on the conclusion	b. Fix: CPU from saved or GPU (A100-h)
E4b controls (X1): first token, f0, layers 24/40/56 only	Does not affect the conclusion. The permuted-gold controls are ≈0 (−0.005 to +0.006) at all three layers.	**The other layers are C fixable**: residuals for al probe layers are stored `probe_controls` can b with the layer filter rem (C12:137), at Stage 13 X (one short CPU job). Fir token and f0 have the s limits as E4b.
E4 lens: first + surname-first tokens only; report shows first only	The surname rows **are already computed** in `361_stage11/summary/E4_titlepos_lens.csv`. They tell the same story as the first-token rows: public-lens target − control at the title position has CIs that include 0 at L16–L52 and is positive from L56 (+0.043) to L62 (+0.055). The first-token rows turn positive from L60. All tokens would need logits that were not stored.	CPU for the surname ro (already in the CSV). All need GPU: about the tit cost, ≈ **2–2.5 A100-h**.
Stage 10: not all 64 layers. public/logit 17 layers; self100 and fitA 6; nothing below L24 except fitA L16 (not run)	Direction unclear for "where does it first appear". Layers 25–31, 33–39 and 41–47 are not sampled; the transition region is covered densely only from L48. The causal E1/E2 curves (patch at 24/32/40/48…, knockout from 0/24/32…) give the timing, so the conclusion does not rest on Stage 10 layer density.	GPU. The public lens co L0–62, so it could be ru Estimate: the main rou 6.3 h for 25 readout-lay over 7 conditions, mos forward cost. Adding th missing layers for publi logit ≈ **8–15 A100-h**.
X1–X3 extras: 6 main layers only (no L50–61); X1_by_format 5 readout-layers; X1_cond_by_first_role 3	Does not affect the conclusion. These are descriptive splits.	**CPU**: the raw data for L exists, so PLAN (X:26, XF XC:87) can be extended takes minutes.
Stage 10 secondary `mean_nlr_surname_first` (one token per author)	By design. It is a sensitivity metric, not the primary one.	–
Stage 11/12 causal sample: 1,186 of 9,108 (E1, E1t, E2, E2n, P1, P2, F2)	Neutral. The sample is balanced within bins, and the CIs at the headline points are narrow (E1 L48 0.807–0.842; E2 L0 0.80–0.86).	GPU, scaling to all 6,84 + control papers (×5.7 2.8 h, E2 ≈ 1.5 h, E2n ≈ P1 ≈ 8.8 h, P2 ≈ 2.7 h, load.
F1: 40 fit papers	Weakens F2's null result: a noisy lens could hide a signal. The F1 lens agrees poorly with the WikiText lenses at L24–48 (cosine 0.27–0.70, handbook E-part).	GPU split-half fit: about per prompt (`alens/progress.json` ≈ 3 h; a larger fit scales prompt count.

Restricted cell	a. Effect on the conclusion	b. Fix: CPU from saved or GPU (A100-h)
N1: analysis restricted to R0/R1; only `mean_nlr` used	By design. first_author_nlr and first_token_nlr are stored, so variants that use only those tokens are CPU-possible.	CPU (stored).
N2 and stage12_cpu B/C: final 63 only	Does not affect the conclusion. The N2 layers were already run post hoc. B and C could use other readouts from `paper_title_contrasts`.	CPU.
R1 per-author: 495 papers (mixed-hit R1 only)	By design: it needs both a written and a missed author in the same paper.	–

Restrictions checked and not found:

- None of these Stage 10–12 analyses uses only the first author. Every per-token experiment scores all authors. The E3 chain smoke check uses the first author only as a check, not as an analysis.
- No mean-of-five analysis silently uses one false title. All `spec` values outside E1, E1t, E2, E2n, P1, P2, E4b and X1 average false_0..4 (S10:863; X:66; XF:58; A11:122, 246; A12:140; C12 via `paper_title_contrasts`).

3. Points I could not verify in code

- **P1 split by token role** (handbook E-part, lines 91–96: initial 0.502/0.917, surname from the name positions 0.567/0.925). The script that produced these numbers is not among the Stage 12 scripts. The handbook says the preview script was not saved and the values were recomputed in a research script. I did not locate or rerun it.
- **E4b confirmatory sample (896 + 499) and Stage 13 X6, X7 and addendum 9.** These are Stage 13 code (`13_stage13/stage13_e4b_group.py` and others). I took them from the Stage 13 reports, not from the code (outside the Part B scope).
- **GPU-hour estimates.** These use unit compute seconds from the logs plus an assumed 0.3–0.5 h model load per job. I did not query `sacct` elapsed times (read-only; slurmctld calls avoided). Treat the estimates as ±30%.
- **The frozen-protocol script hash discrepancy** (U34 in the handbook: freeze sha 847e7dde vs run sha 9107c80b) is open and was not checked here.
- **Handbook vs code line numbers.** The handbook's line references matched the code wherever I compared them (S11:57, 236–251, 254–270, 358, 366–367; A11:56–60, 150–202; C12:111–162). No disagreement was found.

Part C — scoring-unit audit: Stage 13 and post-defence analyses

Read-only audit, 2026-11-05 (evening). Every cell below was checked in code. No files were edited and no jobs were submitted.

Abbreviations used below:

- `s13` = `~/jlens_work/13_stage13/381_stage13.py`
- `s11` = `~/jlens_work/11_stage11/361_stage11.py`
- `r0r1` = `~/jlens_work/10_stage10/stage10_extra_r0r1.py`
- `R13` = `/scratch/zhang.zhiqi2/jlens/381_stage13`
- `R391` = `/scratch/zhang.zhiqi2/jlens/391_groups`

0. Facts used in several rows (verified)

- **"First author"** means `C.ctx[pid][0]`. `s11.contexts_of` (lines 80–86) sorts the contexts by `author_index`, and `author_index` 0 exists for all 9,108 papers (checked in `351_stage10/cohort/name_tokens.parquet`). So index 0 is always the first listed author, never a later author standing in for an excluded one.
- **`name`** = `C.ctx[pid][0][2]`, the full token list of that name: given name or initial, punctuation, and surname tokens.
- **`s13.score_tf`** (lines 200–207) returns the **mean over all of those tokens** of log10(1/rank), read at the final-layer output (`run_final`, hook on layer 63 + unembed, lines 181–197). It uses teacher forcing.
- So X1, X2, X3 and X4 score **the first author, all name tokens, model output L63**. They do not score the first token only.
- **Prompt in X1–X4:** `'Who are the authors of the paper titled "{t}"?\nAuthors:'` (s13 line 48). It has no year, unlike the Stage 10 prompt.
- **Sample sizes:**
- Stage 11 sample: 1,186 papers, 4,307 author contexts, 19,077 name tokens.
- X1–X4 pairs (`config/pairs.parquet`): 1,178 pairs (tune 596 / test 582; target 589 / control 589), 4,271 contexts.
- **First authors are 1,178 / 4,271 = 28% of the author contexts** in the pair papers (3.63 authors per paper).
- **Stage 10 readout layers:** logit, jlens_public, self100 and fitA at L24, 32, 40, 48, 50–62, plus final@63. The same set exists for the true run and for false_0..4 (`351_stage10/raw/`). No analysis uses all 64 layers.

1. Main table (one row per experiment)

exp	1 authors scored	2 tokens scored	3 comparison	4 layers	5 sample
M1 top-1 classes + title sensitivity	all authors	all name tokens, plus a "first token of each name" subset	classes: none (absolute). Sensitivity: **true vs false_0 only** (plus delete for classes)	Stage 10 grid (L24–62 subset + 63)	all 9,108 papers, 149,999 tokens
M2 linear probes	n/a (targets are field / year / log citations / recognised; the docstring mentions a first-author target, but the code has none: lines 212–214)	n/a	true-title run, and **false_0** run (donor labels)	title L16–60 (9 layers); colon L16–60 + L63	9,108
M2b TF-IDF words vs probe	n/a	n/a	n/a (true run, title position)	L16–60	9,108
M2c BERT baseline	n/a	n/a	n/a	BERT layers 0–12 vs Qwen L16–60	9,108
M3 task contrast (identity I, transfer T)	n/a (year / citations / field)	n/a	same paper across the 3 task prompts vs different papers	15 layers [4, 8, 16, 24, 28, 32, 36, 40, 44, 48, 52, 56, 60, 62, 63]	9,108
M3 scores (U46, post-hoc)	**first author only** (year: all year tokens)	**first token only** (`au_ids = [first[0][2][0]]`)	none (absolute; target – control within bin)	output only	9,108
M4 (= Stage 12 F1/F2)	all authors	all name tokens	true vs mean of 5 false	F2 at FIT_LAYERS [24,	Stage 11 sample 1,186

exp	1 authors scored	2 tokens scored	3 comparison	4 layers	5 sample
				32, 40, 44, 48, 52, 56, 62]	
M5 Part A (name-position discrimination)	all authors	all tokens; also roles name_first / surname_first / surname_cont	**all 5 false titles** (paired AUROC)	final@63; logit and public at L24, 32, 40, 48, 50–62	9,108
M5 Part B (title / colon probe, true vs false)	n/a	n/a	**false_0 only**	title L16–60, colon + L63	9,108 (18,216 rows)
X1 cross-task patching	**first author only**	all name tokens (mean)	**one partner B** (corrupt) + one random C (null)	7 layers [4, 8, 16, 24, 32, 40, 48]; conclusions only at ⩾ L24	1,178 pairs (tune 596 / test 582)
X1 similarity	n/a	n/a	same paper / two tasks vs different papers / same task	15 M3 layers	9,108
X1 vs C (V31)	as X1	as X1	B baseline re-expressed against C	as X1	as X1
X2 subspace patching	**first author only** (scoring). **S_auth** is fitted on the first author's **first token** unembedding	all name tokens (scoring)	**one partner B** (carry); μ (remove); 3 random subspaces	5 layers [16, 24, 32, 40, 48]	1,178 pairs (tune 596 / test 582); subspaces fitted on papers outside the pair teams
X3 paraphrase / lexical neighbour D	behaviour: **first author only**. Representations: none	all name tokens (mean)	primary **one D** per pair; secondary **one partner B**	15 layers (s13 line 45); title_last and colon	1,186 generated → 1,137 run → primary 1,115 /

exp	1 authors scored	2 tokens scored	3 comparison	4 layers	5 sample
					sensitivity 1,015
X3 by token role (V51)	first author	per token by role	O / P / B / D	–	**not run**: no `raw/para_tok`; job 11637611 was cancelled on the user's instruction (PROGRESS 18:10)
X4 patchscope	**first author only** (the target prompt asks "The first author of the paper is"). Year: all year tokens; field: all field tokens	all name tokens (mean)	**self vs one partner B**	7 layers [8, 16, 24, 32, 40, 48, 56] at title_last and colon	1,178 pairs
X4 + x4tok (role split)	first author	**per token by role** (given_first / initial_first / surname_first / all)	self vs partner B	**L8 and L40 only**, title position only	1,178 pairs (b7 batch = original)
X4 text baselines	first author	kNN: surname. Supervised: given_first and surname_first tokens	own title vs partner title	n/a	9,108 (team-excluded) / 1,178 pairs
X5 layer order (v1 / v2)	inherited from each source row (mixed)	inherited	inherited	each source's own grid	inherited
X5 retrieval (V56)	n/a	n/a	top-1 among the gallery, 6 task pairs	15 M3 layers	all 9,108; and pairs tune 596 / test 582

exp	1 authors scored	2 tokens scored	3 comparison	4 layers	5 sample
X6 E1 with false_1..4	all authors	all name tokens (paper mean)	**each of the 5 false titles, and their average**	15 layers [24, 32, 40, 48, 50, 52, 54, 56–63]	Stage 11 sample 1,186 (19,077 tokens)
X7 knockout on the false run	all authors	all name tokens	**false_0 only**	13 knockout start layers [0, 24, 32, 40, 48, 50, 52, 54, 56, 58, 60, 62, 64 = none]	Stage 11 sample 1,186
X6 / X7 by role	all authors	per role (name_first, surname_first, surname_cont, …)	as X6 / X7	as X6 / X7	1,186
E4b split / group / trainstd (Stage 11 probe re-runs)	**all authors** (gold = every author's first token, `kind == 'first'`; target = the L63 ':' state)	**first token only**	**false_0 only**	9 layers [16, 24, 32, 40, 44, 48, 52, 56, 60]; controls only at L24 / 40 / 56	9,108 fitted. Confirmatory (team folds + title-reuse dropped) scores only **1,395 target + control papers** (7,245 of 9,108 dropped for title reuse)
Groups P1 / P2 / P3 (addenda 7 / 8; Stage 10 output)	**unwritten authors only** (by design). P2 split into unwritten first / unwritten other. Written authors as a reference	all name tokens (roles also reported)	**mean of 5 false titles**	primary final@63; curves on the full Stage 10 grid	P1 A 3,417 / B 2,148. P2-first C 292 / A 3,199. P2-other C 390 / A 2,826. P3 R0 8,288. Papers with an inconsistent written count dropped
Group regressions	as P1–P3; author-level version with position	all tokens	mean of 5 false	L63	P1 5,565 papers / 19,893 author rows

exp	1 authors scored	2 tokens scored	3 comparison	4 layers	5 sample
Stage 11–13 by group (s1113)	unwritten / unwritten_first / unwritten_other for E1, E2, X6, X7, P1, P2, N2, F2. X1, X3, X4: **papers whose first author was not written**. E4b, N1, M2: all authors (including written ones in C / D)	as the source	as the source (E1 / X7 / P1 / P2: false_0; X6: 5 false)	as the source	**Stage 11 sample only: A 557, B 586, C 36, D 5** (unwritten first: C 26, D 2)
GPU re-runs E1 / E2 (R2, U0)	all authors run; analysis on **unwritten** (primary), written as a reference	all name tokens	**false_0 only** (Stage 11 design)	E1 15 PATCH_LAYERS; E2 13 KO_LAYERS	R2 204 (only **28** papers with unwritten authors → n < 30, descriptive). U0 600 (599 used)
GPU re-runs P1 / P2 (R2 only)	as above	all tokens	false_0 ("none" = unpatched false_0)	P1 PP_LAYERS [32, 40, 44, 48, 50, 52, 56]; P2 PC_LAYERS [8–56, 11 layers]	R2 204 (reference)
num / den splits (E1 / E2, P1 / P2)	unwritten (+ written for C / R2)	all tokens	false_0	as the source	A 557, B 586, C 32, D 5, U0 599, R2 28
All-token probe (addendum 9)	**all authors** (target = mean recorded L63 state over that role's tokens, all authors; unwritten split for the groups)	**all name tokens, by role**: initial, given_first, given_cont, surname_first, surname_cont (punct and	**false_0 only**	9 layers L16–60	fitted on 9,108. Scored on target + control after the title-reuse drop: first / surname_first 1,395, initial

exp	1 authors scored	2 tokens scored	3 comparison	4 layers	5 sample
		particle excluded); plus e4b_parity and first_alltok			1,590, given_first 1,706, surname_cont 1,754
All-token permutation controls	as above	as above	gold of a random other paper / of a random paper in the same field, scored against the same true − false_0 predictions	9 layers	as above

Answer to the specific question. Yes: X1, X2, X3 (behaviour) and X4 all score the first author only (s13 lines 215, 317, 239; stage13_x2.py line 84).

- They score all of that author's name tokens, not just the first token.
- They read the model output L63.
- They compare against one partner B (X3 primary: one lexical neighbour D).
- X2's S_auth subspace is additionally built from the first author's first token only (x2_fit line 92).
- X1 similarity, and the X3 representation analysis, score no author at all.

2. Restricted cells: effect, how to fix, and what to do

GPU estimates are scaled from measured A100-80GB unit times:

- **Stage 13 GPU-log totals** (sums of the per-unit times in the logs): xpatch 16 min / 24 shards; pscope 23 min; para_run 9 min; task 32 min / 72; xf_patch 74 s per shard; ko_false 46 s per shard.
- **sacct:** X2 11655027 15 min for 10 shards (about 35 min for 24); all-token GPU 11655112 / 130 / 136 about 41 min for 72 shards; addendum-8 re-runs about 75 min total.
- **Added to every estimate:** about 10–15 min of model loading per job.
- **First-author → all-author scale-up:** × 3.6 on the scoring passes.

#	restricted cell	a. effect on the conclusion	b. CPU or GPU fix (what is stored estimate
1	X1: first author only (28% of contexts)	**Unclear, probably small.** The task-independence verdict compares sources on the same pairs. Nothing in the method makes it author-specific. But author position changes the signal size (addendum 8), so the levels of R may differ.	**GPU.** `raw/xpatch` stores one mea first-author score per pair. No residuals or other authors. About A100-h (16 min × 3.6 + load).
2	X1 / X2 / X4: one partner B per pair	**Unclear (noise, not bias).** B is matched on field and length. X1 also has a random C, and the C-based normalisation (x1_vsC) gives the same verdicts.	GPU (new corrupt runs per extra partner): about 0.3 h per extra par for X1.
3	X2: first-author scoring + S_auth from the **first token** (73% initials)	**Possibly larger.** An "author" subspace aimed at the first-token unembedding (mostly initials) may miss the surname information. The S_auth decoder is already weak (held-out $R^2 \leqslant 0.003$), so H3 may be under-tested. The H2 verdict (S_id does not carry) is not affected.	**Partly CPU:** S_auth can be refitte CPU with surname_first targets (M residual cache + W_U are on disk; x2_fit route). The carry / remove a need **GPU**: about 0.6–0.7 A100-h (author, new subspace), about 2 h all authors.
4	X3 behaviour: first author only; one D	**Small for the conclusion.** The X3 conclusion rests on representations (author-free) and on the text baseline. Behaviour (para − D, recovery 0.90) is secondary.	**GPU:** para_run stores the per-pair first-author mean only. About 0.5- A100-h for all authors. Per-token X (V51) also needs GPU and was cancelled by the user.
5	X4: first author only	**No effect on the stated question.** The target prompt asks for "the first author". Scoring others would need a different target prompt, which is a design change.	GPU plus a new design (needs approval).
6	X4 role split (x4tok): only L8 and L40, title position only	**Unclear for other layers.** The role claim ("L8 author signal mainly on the surname") is made at L8 only.	GPU: about 0.3 A100-h for the remaining 5 layers + colon (pscop min total; author part only).
7	M3 scores: first author, **first token only**	**Smaller than an all-token score would show.** Initials carry little title-specific signal (Stage 10 role table); target − control +0.147 is descriptive only.	GPU (task runs stored only this sc about 0.6–0.8 A100-h. Not worth it Stage 10 has all-token output sco for all authors (with-year prompt)
8	E4b (all variants): **first token only**	**Smaller (understated).** Addendum 9 (pre-registered version V0, false_0 only; 1,395 / 1,706 papers) shows surname_first L40 +0.071 and given_first +0.124, against first-token +0.061.	Done: addendum 9 (all-token) exis

#	restricted cell	a. effect on the conclusion	b. CPU or GPU fix (what is stored estimate
9	E4b / all-token probe / M2 / M5-B: **false_0 only**	**Unclear.** X6 shows the timing is the same for all 5 false titles (within 0.02), so a large change is unlikely. But a single false title adds donor-specific noise.	**GPU:** Stage 11 titlepos stored resi for true and false_0 only (false_1.. files have lens ranks, no residuals; verified). Prompt-only forward wi residuals, 4 conditions × 72 shard about 2–3 A100-h (scaled from the token job, about 41 min / 72 shard
10	E4b confirmatory / all-token probe: **title-reuse exclusion drops 80% of papers** (scores 1,395 target + control papers; 7,245 of 9,108 flagged)	**Unclear.** The CIs are wider. The included papers are those whose false_0 donor fell in the same fold, so the sample is not random.	**CPU:** per-paper spec for all 9,108 papers (dropped flag included) is stored in `summary_cpu_trainstd/perpaper be_spec.parquet` and `summary_alltok/probe/perpaper arquet`. The "not dropped" config already exists for E4b (4 configs). F all-token, a reuse-included sensit takes minutes on CPU.
11	All-token permutation controls (new)	Preliminary (my plain means, target + control, title reuse dropped; **not team-weighted, not final**): random-gold controls are about 0 for e4b_parity / first / surname_first / surname_cont (⩽ 0.016). For **initial** they are +0.018 to +0.021 at L40 / L56, against a real +0.044 / +0.054.	CPU, done; the main session is summarising.
12	X7: **false_0 only**	**Could matter for the pass / fail call.** Control S(L52) = 0.051 misses the 0.05 threshold by 0.001, so averaging 5 false titles could flip it. X6 suggests little spread.	**GPU:** ko_false for false_1..4, 24 sh × 4 × 46 s ≈ 75 min + load ≈ 1.3 A100-h.
13	M1 title sensitivity: true vs false_0 only	No effect (descriptive).	**CPU:** `cond_false_1..4` raw files l `top1` (verified). Minutes.
14	Groups P1–P3: unwritten authors only	By design (circularity rule). Not a defect; the written reference exists.	–
15	GPU re-runs / num-den: false_0 only; R2 unwritten n = 28	Same as Stage 11 design (X6 covers the 5 titles for A / B). R2 is a reference only; n < 30 is stated.	GPU for more false titles (about 0. per title for 804 papers); R2 canno gain unwritten authors.
16	s1113: Stage 11–13 measures by group use the Stage 11 sample, so **C 36 / D 5** papers (C first-unwritten 26, D 2); X1 / X3 / X4 need a non-written first author	**C and D conclusions are not possible** (n < 30, descriptive). The primary P2 (Stage 10, full cohort) is unaffected.	GPU: E1 + E2 for the remaining ab 475 C papers ≈ 0.7–1 A100-h (scal from the re-runs). X1 / X3 / X4 for C would need new pairs and about 1 h more.

#	restricted cell	a. effect on the conclusion	b. CPU or GPU fix (what is stored estimate
17	s1113 E4b by group includes written authors in C / D	Small (C / D are descriptive anyway).	Superseded by addendum 9 `summary_alltok/groups_unwritty_role.csv` (per-role unwritten).
18	Layers: no analysis uses all 64 layers (grids of 5–17 layers)	No effect on conclusions: the curves are dense where it matters (L48–63). X1 stops at L48 (R = 0.42 there, about 0 expected later per E1).	–

3. M5 and V42: status

M5 (V29)

- **Status:** done on CPU, post-hoc (there is no pre-registered rule). Job 11637638, 2026-11-05 01:53.
- **Files:** `R13/reports/m5_report.md`, `summary_m5/m5_name_discrimination.csv`, `m5_titlepos_probe.csv`.
- **Coverage map:**
- M5-2 patchscope → covered by X4.
- M5-1 probes at title / ':' → covered by M2 / E4b / M3.
- Probes and patchscope **at the name positions** → **not done**. They need GPU because no name-position residuals were ever saved. The script estimates about 24 short A100 units for the 1,186-paper sample, true + false_0 (about 2 GB per condition).
- **Part A** (lens true / false discrimination at the name positions; all authors, all tokens, 5 false titles, team CI):
- final@63: 0.919 [0.915, 0.924] (target 0.976, control 0.812; target − control within bin +0.140 [+0.128, +0.152]).
- logit lens: about 0.51–0.53 at L24–L48, then a rise from L50 (0.563) to 0.899 at L62.
- public J-lens: 0.52–0.56 at L24–L53 (target − control negative at L48–55, e.g. L48 −0.057), 0.902 at L62.
- **Part B** (probe, true vs false_0 at title / ':'): AUROC 0.501–0.505 at the title and 0.506–0.516 at ':', which is chance.
- **In report v28: yes.** Part 5 "CPU 补充 → M5 能用已有数据回答的部分（V29，事后）" has the Part A and Part B numbers, the GPU-needed note and the overview line. Report text line 179 also mentions it, and Part IV (M5③) explains why.
- **Stale:** `STATUS.md` still says "13 M5 | dropped / merged into X4".

V42 (paper bootstrap vs team bootstrap; X4 recognition on tune / test)

- **Status:** done on CPU, post-hoc. Jobs 11637483 (M3 paper bootstrap, 63 min) and 11637554 (44 s).

- **Files:** `R13/reports/v42_paper_vs_team.md`, `summary_v42/v42_paper_vs_team.csv` (1,155 rows), `summary_v42/x4_recognition_split.csv`, `summary_v42/m3_paper_vs_team.csv`.
- **Results:**
- Width ratio (paper / team): median 0.97–1.00 (range 0.71–1.09).
- "Interval excludes 0" changes in only 2 rows: X4 test D_year_title L16 (+0.0044) and X7 target T L60 (+0.0029).
- X4 recognition: the layers chosen on tune are **not confirmed on test** (author · title L8: +0.0231 [−0.0006, +0.0478]). Test reaches > 0 from L16 to L48 at the title (e.g. L40 +0.060 [+0.035, +0.084]).
- **In report v28: yes.** Part 5 has "论文 bootstrap 与团队聚类区间的对照（V42，事后）" with all of these numbers.

Related (V56, U46, V21): also in v28 Part 5 (X5 retrieval, M3 scores, N2 by layer).

All-token probe (addendum 9):

- v28 still says "尚无结果" (Part 5 "进行中" and the E4b caveat boxes at text lines 179, 455, 1173, 1248). The results now exist (`reports/e4b_alltok.md`, PROGRESS 17:55), and so do the permutation controls (63 files).
- **Not yet in any document.**

4. Not verifiable / caveats

- **Written flag.** `groups_build.py` line 41 assigns `author_index` by `cumcount` over the reconstructed `hits` list. I did not verify that `hits` follows the Stage 10 author order for every paper. It is inherited from `stage10_extra_r0r1.match()`, and the rerun_counts consistency filter drops mismatching papers.
- **GPU hour estimates** are scalings of the measured unit times. They are not measured for the new configurations. The titlepos per-condition time is inferred from the all-token job, because I found no Stage 11 titlepos timing lines.
- **X2 analysis estimator lines** (`stage13_x2_analyze.py`) were not read line by line. The scoring unit was verified in `stage13_x2.py` and `stage13_x2_fit.py`. The verdict numbers are quoted from PROGRESS / the X2 entry, not recomputed.
- **M4 / F2** was only checked at the docstring and layer level here (Part B of the audit covers Stage 12).
- **Docstring vs code.** The `stage13_cpu.py` docstring lists a "first-author first token" M2 target, but the code (lines 212–214) has none. The M5 report's coverage table cites "first-author token … M2" accordingly, and that is really E4b, not M2.
- **STATUS.md is stale** for X2 (marked "not done", but it was done 2026-11-05) and for M5.
- **Item 11 control numbers** are my unweighted plain means for orientation only. The main session's summary (team CI) is authoritative.